

**MTH166**

**Lecture-6**

**Clairaut's Equation**

## **Topic:**

Clairaut's Equation: A special case of equations solvable for  $y$

## **Learning Outcomes:**

1. How to find general solution (Not singular solution) of Clairaut's equation.
2. How to solve equations solvable for  $y$  i.e.  $y = f(x, p)$

## Clairaut's Equation:

An equation of the form:  $y = px \pm f(p)$  (1)

is called Clairaut's equation where  $p = \frac{dy}{dx}$

General solution of equation (1) is given by:

$y = cx \pm f(c)$  (Just replace  $p$  by  $c$  in the question, we get the answer)

\* It is also called as a special case of equations solvable for  $y$ :  $y = f(x, p)$

\*\* It is a very good concept for asking MCQ

**Find the general solution of following differential equations:**

**Problem 1.**       $xp^2 - yp + a = 0$  (1)

**Solution:**       $xp^2 - yp + a = 0$

$$\Rightarrow yp = xp^2 + a$$

$$\Rightarrow y = px + \frac{a}{p} \quad (2)$$

Which is of Clairaut's form:  $y = px + f(p)$

So, the general solution of equation (1) is given by putting  $p = c$  in equation (2)

i.e.  $y = cx + \frac{a}{c}$  **Answer.**

**Problem 2.**      $p = \log(px - y)$  (1)

**Solution:**      $p = \log(px - y)$

$$\Rightarrow e^p = e^{\log(px-y)}$$

$$\Rightarrow e^p = (px - y)$$

$$\Rightarrow y = px - e^p$$
 (2)

Which is of Clairaut's form:  $y = px \pm f(p)$

So, the general solution of equation (1) is given by putting  $p = c$  in equation (2)

i.e.      $y = cx - e^c$      **Answer.**

**Problem 3.**       $y = px + \sqrt{a^2p^2 + b^2}$  **(1)**

**Solution:**       $y = px + \sqrt{a^2p^2 + b^2}$

Which is of Clairaut's form:  $y = px + f(p)$

So, the general solution of equation (1) is given by putting  $p = c$  in equation (1)

i.e.     $y = cx + \sqrt{a^2c^2 + b^2}$       **Answer.**

**Problem 4.**       $\sin px \cos y = \cos px \sin y + p$  (1)

**Solution:**       $\sin px \cos y = \cos px \sin y + p$

$$\Rightarrow \sin px \cos y - \cos px \sin y = p$$

$$\Rightarrow \sin(px - y) = p$$

$$\Rightarrow (px - y) = \sin^{-1}p$$

$$\Rightarrow y = px - \sin^{-1}p \tag{2}$$

Which is of Clairaut's form:  $y = px \pm f(p)$

So, the general solution of equation (1) is given by putting  $p = c$  in equation (2)

i.e.  $y = cx - \sin^{-1}c$  **Answer.**

**Problem 5.**  $y + 2 \left( \frac{dy}{dx} \right)^2 = (x + 1) \left( \frac{dy}{dx} \right)$  (1)

**Solution:**  $y + 2p^2 = (x + 1)p$

$$\Rightarrow y = px + p - 2p^2$$

$$\Rightarrow y = px + p(1 - 2p) \quad (2)$$

Which is of Clairaut's form:  $y = px + f(p)$

So, the general solution of equation (1) is given by putting  $p = c$  in equation (2)

i.e.  $y = cx + c(1 - 2c)$  **Answer.**



**Problem 6.**  $(y - px)(p - 1) = p$  (1)

**Solution:**  $(y - px)(p - 1) = p$

$$\Rightarrow y - px = \frac{p}{p-1}$$

$$\Rightarrow y = px + \frac{p}{p-1} \quad (2)$$

Which is of Clairaut's form:  $y = px + f(p)$

So, the general solution of equation (1) is given by putting  $p = c$  in equation (2)

i.e.  $y = cx + \frac{c}{c-1}$  **Answer.**

## Polling Quiz

Which of the following is I.F. for equation:  $(x dy - y dx) + a(x^2 + y^2) dx = 0$

(A)  $\frac{1}{x^2}$

(B)  $\frac{1}{y^2}$

(C)  $\frac{1}{xy}$

(D)  $\frac{1}{(x^2 + y^2)}$

**Equations solvable for y:**  $y = f(x, p)$

**Problem.**  $y = 2px + p^n$  (1)

**Solution:**  $y = 2px + p^n$  (1)

Equation (1) is neither of Clairaut's form:  $y = px + f(p)$  nor it can be solved for p by factorisation.

But equation (1) is of the form  $y = f(x, p)$  i.e. solvable for y. So, Differentiating equation (1) w.r.t x, we get:

$$y = 2px + p^n \quad (1)$$

$$\Rightarrow \frac{dy}{dx} = 2 \left[ p \frac{d}{dx}(x) + x \frac{d}{dx}(p) \right] + np^{n-1} \frac{d}{dx}(p)$$

$$\Rightarrow p = 2 \left[ p + x \frac{dp}{dx} \right] + np^{n-1} \frac{dp}{dx}$$

$$\Rightarrow p + 2x \frac{dp}{dx} + np^{n-1} \frac{dp}{dx} = 0$$

$$\Rightarrow p + (2x + np^{n-1}) \frac{dp}{dx} = 0$$

$$\Rightarrow p dx + (2x + np^{n-1}) dp = 0 \quad (2)$$

Comparing it with:  $M dx + N dp = 0$

$$\text{Here } M = p \Rightarrow \frac{\partial M}{\partial p} = 1$$

$$\text{and } N = (2x + np^{n-1}) \Rightarrow \frac{\partial N}{\partial x} = 2$$

Since,  $\frac{\partial M}{\partial p} \neq \frac{\partial N}{\partial x}$ , so, equation (2) is non-exact.

$$\text{Here } \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial p}}{M} = \frac{2-1}{p} = \frac{1}{p} = f(p)$$

$$\text{So, } I.F. = e^{\int f(p)dp} = e^{\int \frac{1}{p} dp} = e^{\log p} = p$$

Multiplying equation (2) by I.F.

$$[pdx + (2x + np^{n-1})dp] \times p = 0$$

$$\Rightarrow p^2 dx + (2xp + np^n)dp = 0$$

Which is an exact differential equation.

$$\text{Solution: } \int_{p=\text{con.}} Mdx + \int (\text{Terms of } N \text{ free from } x)dp = c$$

$$\Rightarrow \int_{p=\text{con.}} p^2 dx + \int (np^n)dp = c$$

$$\Rightarrow p^2 x + n \frac{p^{n+1}}{n+1} = c$$

$$\Rightarrow x = \frac{c}{p^2} - n \frac{p^{n-1}}{n+1} \quad (3)$$

$$\Rightarrow x = \frac{c}{p^2} - n \frac{p^{n-1}}{n+1} \quad (3)$$

Substituting this value of x from equation (3) in equation (1), we get:

$$y = 2px + p^n$$

$$\Rightarrow y = 2p \left( \frac{c}{p^2} - n \frac{p^{n-1}}{n+1} \right) + p^n \quad (4)$$

Thus, equations (3) and (4) taken together, in parameter p, give the general solution of equation (1).

## Polling Quiz

**Solution of differential equation  $xdy - ydx - 2y^3dy = 0$  is:**

(A)  $x + y^3 = c$

(B)  $x + y^3 - cy = 0$

(C)  $x + y^3 + cy = 0$

(D) None of these

**Equations solvable for y:**  $x = f(y, p)$

**Problem.**  $x = \frac{y - y^2 p^3}{2p}$  (1)

*Try it yourself.*



